NOT MEASUREMENT SENSITIVE

MIL-T-5882D
20 July 1989
SUPERSEDING
MIL-T-5882C
8 AUGUST 1978

MILITARY SPECIFICATION

TRANSMITTER, PRESSURE, MULTIPURPOSE,
0-50 PSI, MS28005-1

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of remote indicating, 320 degrees movement synchro style, general purpose pressure gage transmitter.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

2.1.1 Specification and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEOR, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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SPECIFICATIONS

FEDERAL

PP-B-601	Boxes, Wood, Cleated Plywood
PP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation-Packaging, Methods Of
MIL-T-5350	Transmitters, Synchro Operated, Aircraft General Specification For
MIL-T-7085	Indicator, Pressure, Multipurpose, 0-50 Psi
MIL-B-27497	Bearings, Jewel, Sapphire, Synthetic

STANDARDS

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U. S. Military Property
MIL-STD-781	Reliability Testing For Engineering Development, Qualification & Production
MIL-STD-810C	Environmental Test Methods & Engineering Guidelines
MIL-STD-2073-1	DoD Material Procedures For Development And Application Of Packaging Requirements
MS28005	Transmitter, Pressure-Synchro, Aircraft
MS28010	Indicators, Pressure Synchro, Single, 2-Inch Size

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The transmitter furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (See 4.2 and 6.4).

3.2 General requirements. The requirements of MIL-T-5350 are applicable as requirements of this specification. Additional requirements shall be as specified herein. Where the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.2.1 Reclaimed materials. Reclaimed materials shall be utilized to the maximum extent possible within the quality limits required by this document.

3.3. Design and construction.

3.3.1 General Design. The transmitters shall be designed for use with an MS28010-1 indicator conforming to MIL-I-7085.

3.3.2 Hermetically sealed case. The case shall provide a hermetically sealed enclosure for the entire internal mechanism of the transmitter. The design of the case shall permit removal and replacement of the internal mechanism and resealing of the case. This shall be accomplished without the use of special tools and fixtures unless they are approved by the contracting activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any environment to which the transmitter may be subjected.

3.3.3 Outline dimensions. The outline dimensions of the transmitter shall conform to MS28005.

3.3.4 Bosses. The bosses, as shown on MS28005, shall be provided for external pressure and vent connections to the pressure sensitive element and the opposing pressure element, respectively. The bosses shall be identified with white lettering "P" and "V" permanently inscribed 0.125 inch in height.

3.3.5 Operation. The transmitter shall be capable of operating in any position.

3.3.6 Overpressure stop. An overpressure stop shall be provided and shall be designed to restrain the prime mover. The stop shall allow the pointer of an indicator to travel beyond full scale to a point approximately midway between the last and the zero graduation.

3.3.7 Pressure connections. The pressure transmitter shall have two pressure connections (external pressure connection and vent connection). The transmitter shall be so designed and constructed that it will be actuated by the difference between the pressure applied to the external pressure connection and the pressure applied to the vent connection, and the synchro signal output shall represent external pressure minus vent pressure. The angular position of the rotor of the synchro relative to the stator shall be essentially a linear function of external pressure minus vent pressure.

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3.3.8 Silicone oil. The transmitter shall not contain silicone oil.

3.4 Bearings synthetic. When sapphire or ruby jewel synthetic bearings are used they shall be in accordance with MIL-B-27497.

3.5 Weight. The weight of the transmitter shall not exceed 1.4 pounds.

3.6 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.1.1 General. The transmitter shall be subjected to applicable tests in accordance with MIL-T-5350 and as specified herein.

4.2 Classification of inspections. The inspection and testing of transmitter shall be classified as follows:

a. Qualification inspection	See 4.3
b. Quality conformance inspection	See 4.4

4.3 Qualification inspection. Qualification tests shall consists of all tests under paragraph 4.5, except 4.5.24, of MIL-T-5350, and the tests of this specification.

4.4 Quality conformance inspection. Inspection shall consist of the following:

a. Individual inspection	See 4.4.1
b. Sampling inspection	See 4.4.2

4.4.1 Individual inspections. Each transmitter submitted for acceptance under contract shall be subjected to the individual inspections specified in MIL-T-5350 and all tests specified in 4.6.

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TABLE I.

MULTI-PURPOSE PRESSURE TRANSMITTER
SCALE ERROR TOLERANCES, ± PSI

TEST CONDITION	TEST POINTS (PSI)						
	0	10	20	25	30	40	50
Room Temperature	0.75	0.75	0.50	0.50	0.50	0.50	1.00
Low Temperature, Operational	1.25	1.25	1.25	1.25	1.25	1.50	1.50
High Temperature, Operational	1.25	1.25	1.25	1.25	1.25	1.50	1.50
Vibration (Post Test Operation)	1.00	1.00	1.00	1.00	1.00	1.00	1.25
Low Temperature Exposure (Operational)	1.25	1.25	1.25	1.25	1.25	1.50	1.50
Room Temp. following Low Temp. Exposure *	0.75	0.75	0.50	0.50	0.50	0.50	1.00
Room Temp. following High Temp. Exposure *	0.75	0.75	0.50	0.50	0.50	0.50	1.00
Seasoning	1.00	1.00	1.00	1.00	1.00	1.00	1.25
Endurance	1.25	1.25	1.25	1.25	1.25	1.50	1.50
Power Supply Variation *	0.75	0.75	0.50	0.50	0.50	0.50	1.00
Salt Spray *	0.75	0.75	0.50	0.50	0.50	0.50	1.00
Suction and Overpressure	1.50	1.50	1.50	1.50	1.50	1.50	1.50
Shock (Post Test Operation)*	0.75	0.75	0.50	0.50	0.50	0.50	1.00

*NOTE: The tolerances shown are applicable unless allowable shifts in calibration have occurred during preceeding tests, in which case the tolerance of such preceeding test shall apply.

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4.4.2 Sampling. Sampling shall be as specified in MIL-T-5350.

4.5 Inspection conditions. Unless otherwise specified, the transmitter shall be tested in a horizontal position with the pressure connection at the top.

4.6 Test Methods.

4.6.1 Scale error. The test points and tolerances for the error test shall be as specified in table I.

4.6.2 Friction. Tolerances for the friction test shall be as specified in table II.

TABLE II. PRESSURE TRANSMITTER.

Test	Tolerance ± psi
Vibration:	
Pointer oscillation	0.50
Friction:	
Room temperature	0.50
High temperature	0.75
Low temperature	1.00
Position error at 30 psi	0.50
Leak	0.10

4.6.3 Sealing. Pressure and vent connections shall be closed with suitable plugs for this test. Each transmitter shall be immersed in a suitable deaerated liquid, such as water, within an enclosure. The absolute pressure of the air above the liquid shall be reduced to 2.5 inches Hg. Any bubbles coming from within the transmitter case shall be considered as leakage and shall be cause for rejection. Bubbles which are the result of entrapped air in the various exterior parts of the case shall not be considered as a leak. A wetting agent shall be used in the liquid to facilitate removal of surface bubbles.

4.6.3.1 The sealing test of MIL-T-5350 may be used as an alternate method.

4.6.4 Leak. A pressure of 50 pounds per square inch shall be applied to the pressure connection and a pressure of 20 pounds per square inch shall be applied to the vent connection. The connecting tubes shall be completely sealed at a point within 2 inches of each connection. During a period of one minute, the pointer shall not change its position by more than the tolerance specified in table II. This test may be combined with that of the scale error test.

4.6.5 Damping. A pressure sufficient to produce full scale test pointer deflection shall be applied to the transmitter. A quick release valve shall be used to release the pressure. The time interval for the change from 50 psi to 5 psi indication shall be 9 ± 3 seconds. The test shall be conducted with lubricating oil with a viscosity of approximately 2,500 centistokes, such as grade 1005 oil at -65°F, or equivalent. The test shall be conducted at 77°F.

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4.6.6 Suction and overpressure. The pressure port of the transmitter shall be subjected to a pressure of 5 ± 0.5 inches Hg absolute for a period of one minute. The pressure port shall then be subjected to a pressure of 100 psi for a period of ten minutes. Not less than 1 hour following the overpressure, the transmitter shall be tested for scale errors as specified in the individual tests on increasing pressures only. The scale error shall not exceed the tolerances specified in Table I.

4.6.7 Salt fog. After exposure to the specified salt fog test the transmitter shall be examined for corrosion and subjected to the sealing test. No scale or friction error test shall be required.

4.6.8 Power supply variation. Power supply variation tests shall allow fluctuations of +/-4 volts and +/-20 percent of the nominal frequency.

4.6.9 Reliability tests. The reliability test units shall be subjected to the following combined environmental conditions:

a. Temperature: -55°C to +121°C or remotely mounted transmitters or -55° C to +190° C for direct engine mounted transmitters.

b. Temperature cycling: Each temperature cycle shall be 8 hours in length. Following a room temperature dwell of 30 minutes, the chamber temperature will be dropped over a period of 15 minutes to -55° C. Following a 3 hour, -55°C soak, the chamber temperature shall be raised over a 45 minute period to the upper temperature range (+121°C or +190°C). The transmitter shall remain at the high temperature for a period of 3 hours after which the temperature shall be lowered to room temperature over a period of 30 minutes. This constitutes one cycle. This cycle shall continue throughout the duration of the reliability tests, except for moments of equipment shut down to assess transmitter performance.

c. Transmitter on-off cycling: The transmitters shall be off for 15 minutes out of each high temperature and low temperature soak as well as during the 30 minute room temperature period. The transmitters shall be on at all other times.

d. Pressure oscillations: The transmitters shall be subjected to input line pressure oscillations at a frequency of 1 Hz from 20+/-1 to 30+/-1 psi. Pressure oscillations shall occur whenever the transmitters are powered on. When the transmitters are powered off, the input line pressure shall be 0 psi.

e. Vibration: Throughout the pressure cycling, for 10 minutes of each hour the transmitters shall be subjected to sinusoidal vibration cycling in accordance with MIL-STD-810C, method 514.2, using curve D for shock mounted transmitters or curve L for direct engine mounted transmitters. At the end of 4 test cycles (32 hours) during the room temperature phase, reorient the transmitters to another axis and continue testing.

f. Input voltage: Nominal specified voltage.

At the end of each 4 cycles, when the transmitters are being reoriented on the vibration stand, the transmitters shall be scale error tested. Any transmitter which does not stay within the scale error tolerances for endurance in Table I shall be considered a failure. No preventative maintenance shall be performed. Lower test MTBF (Ō) shall not be less than 5000 hours. Accept-reject criteria shall be in accordance with MIL-STD-781, test plan I.

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4.6.10 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with section 4, MIL-P-116, to insure compliance with section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Transmitter shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Transmitter shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation Application. Preservative shall not be used.

5.1.1.4 Unit Packaging. Unless otherwise specified by the contracting activity, each transmitter shall be packaged in quantity unit packs of one each in accordance with Method 1c-1 of MIL-P-116. Each transmitter shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the transmitter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each transmitter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of the transmitter shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged transmitter shall be packed in accordance with ASTM D3951.

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5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The multipurpose pressure transmitter covered by this specification is intended for use to transmit the indicated pressure in the fuel, lubricating oil, and water injection systems of aircraft engines to an applicable indicator.

6.1.1 General. The instruction contained in section 6 of MIL-T-5350 are applicable to and form a part of this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Qualifications. With respect to products requiring qualification, awards will be made only for such products as have prior to time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is OC-ALC/MMEOR, Tinker AFB, OK 73145-5990 and information pertaining to qualification of products may be obtained from that activity.

6.4 Subject term (key word) listing.
Multipurpose
Pressure
Transmitter
0-50 PSI

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
AIR FORCE - 99
Navy -AS

Preparing Activity:
AIR-FORCE - 71

Project number:
6620-0475

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-5882D	Transmitter, Pressure, Multipurpose 0-50 PSI

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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